Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.51_2)

Metric

$$ds^2 = -\mathrm{d}t \otimes \mathrm{d}t + V\left(t, z, r, \phi\right)^2 \mathrm{d}z \otimes \mathrm{d}z + \left(-\frac{F\left(t\right)^2}{r^2 - 1}\right) \mathrm{d}r \otimes \mathrm{d}r + r^2 F\left(t\right)^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, z, r, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}}\right) dt + \left(-\sqrt{\frac{1}{2}} V\left(t, z, r, \phi\right)\right) dz$$

$$n = \left(\sqrt{\frac{1}{2}}\right) dt + \left(\sqrt{\frac{1}{2}} V\left(t, z, r, \phi\right)\right) dz$$

$$m = \left(\frac{\sqrt{\frac{1}{2}} F\left(t\right)}{\sqrt{-r^2 + 1}}\right) dr + \left(i \sqrt{\frac{1}{2}} r F\left(t\right)\right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{\frac{1}{2}} F\left(t\right)}{\sqrt{-r^2 + 1}}\right) dr + \left(-i \sqrt{\frac{1}{2}} r F\left(t\right)\right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(-\frac{\sqrt{2}}{2\,V\left(t,z,r,\phi\right)}\right) dz$$

$$n = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{\sqrt{2}}{2\,V\left(t,z,r,\phi\right)}\right) dz$$

$$m = \left(\frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}}{2\,F\left(t\right)}\right) dr + \left(\frac{i\sqrt{2}}{2\,rF\left(t\right)}\right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}}{2\,F\left(t\right)}\right) dr + \left(-\frac{i\sqrt{2}}{2\,rF\left(t\right)}\right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\,\partial_t - \frac{\sqrt{2}}{2\,V\left(t,z,r,\phi\right)}\,\partial_z$$

$$\Delta = -\frac{1}{2}\sqrt{2}\,\partial_t + \frac{\sqrt{2}}{2\,V\left(t,z,r,\phi\right)}\,\partial_z$$

$$\delta = \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}}{2\,F\left(t\right)}\,\partial_r + \frac{i\sqrt{2}}{2\,rF\left(t\right)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}}{2\,F\left(t\right)}\,\partial_r - \frac{i\sqrt{2}}{2\,rF\left(t\right)}\,\partial_\phi$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{2\,F(t)}$$

$$\kappa = \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} + \frac{i\,\sqrt{2}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)}$$

$$\mu = 0$$

$$\lambda = \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{2\,F(t)}$$

$$\nu = -\frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} + \frac{i\,\sqrt{2}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = -\frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} - \frac{i\,\sqrt{2}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)}$$

$$\pi = \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} - \frac{i\,\sqrt{2}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)}$$

$$\epsilon = -\frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)}{4\,V(t,z,r,\phi)}$$

$$\gamma = \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)}{4\,V(t,z,r,\phi)}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{\frac{\partial^2}{(\partial t)^2} F(t)}{2\,F(t)} + \frac{\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial t} V(t,z,r,\phi)}{2\,F(t)\,V(t,z,r,\phi)} + \frac{r^2\,\frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} + \frac{r\,\frac{\partial}{\partial r} V(t,z,r,\phi)}{2\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial}{\partial r} V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)}$$

$$\Phi_{01} = \frac{\sqrt{r+1}\sqrt{r-1}\,\frac{\partial^2}{\partial t\partial r} V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{r+1}\sqrt{r-1}\,\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial r} V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} + \frac{i\,\frac{\partial^2}{\partial t\partial \phi} V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)} - \frac{i\,\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial \phi} V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)}$$

$$\Phi_{02} = -\frac{\frac{\partial}{\partial t} F(t)^2}{4\,F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2} V(t,z,r,\phi)}{4\,V(t,z,r,\phi)} - \frac{1}{4\,F(t)^2} - \frac{i\,\sqrt{r+1}\sqrt{r-1}\,\frac{\partial^2}{\partial r\partial \phi} V(t,z,r,\phi)}{2\,rF(t)^2\,V(t,z,r,\phi)} + \frac{i\,\sqrt{r+1}\sqrt{r-1}\,\frac{\partial}{\partial \phi} V(t,z,r,\phi)}{2\,r^2F(t)^2\,V(t,z,r,\phi)}$$

$$\Phi_{10} = \frac{\sqrt{r+1}\sqrt{r-1}\,\frac{\partial^2}{\partial t\partial r} V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{r+1}\sqrt{r-1}\,\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial r} V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{i\,\frac{\partial^2}{\partial t\partial \phi} V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)} + \frac{i\,\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial \phi} V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)}$$

$$\Phi_{11} = \frac{\frac{\partial}{\partial t} F(t)^2}{8\,F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} V(t,z,r,\phi)}{8\,V(t,z,r,\phi)} - \frac{r^2\,\frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi)}{8\,F(t)^2\,V(t,z,r,\phi)} + \frac{1}{8\,F(t)^2} + \frac{\frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi)}{8\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial}{\partial r} V(t,z,r,\phi)}{8\,rF(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t,z,r,\phi)}{8\,r^2F(t)^2\,V(t,z,r,\phi)}$$

$$\Phi_{12} = \frac{\sqrt{r+1}\sqrt{r-1}\,\frac{\partial^2}{\partial t\partial r} V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{r+1}\sqrt{r-1}\,\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial r} V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} + \frac{i\,\frac{\partial^2}{\partial t\partial \phi} V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)} - \frac{i\,\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial \phi} V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)}$$

$$\Phi_{20} = -\frac{\frac{\partial}{\partial t} F(t)^2}{4\,F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2} V(t,z,r,\phi)}{4\,V(t,z,r,\phi)} - \frac{1}{4\,F(t)^2} + \frac{i\,\sqrt{r+1}\sqrt{r-1}\,\frac{\partial^2}{\partial r\partial \phi} V(t,z,r,\phi)}{2\,rF(t)^2\,V(t,z,r,\phi)} - \frac{i\,\sqrt{r+1}\sqrt{r-1}\,\frac{\partial}{\partial \phi} V(t,z,r,\phi)}{2\,r^2F(t)^2\,V(t,z,r,\phi)}$$

$$\Phi_{21} = \frac{\sqrt{r+1}\sqrt{r-1}\,\frac{\partial^2}{\partial t\partial r} V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{r+1}\sqrt{r-1}\,\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial r} V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{i\,\frac{\partial^2}{\partial t\partial \phi} V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)} + \frac{i\,\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial \phi} V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)}$$

$$\Phi_{22} = -\frac{\frac{\partial^2}{(\partial t)^2} F(t)}{2\,F(t)} + \frac{\frac{\partial}{\partial t} F(t)\,\frac{\partial}{\partial t} V(t,z,r,\phi)}{2\,F(t)\,V(t,z,r,\phi)} + \frac{r^2\,\frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} + \frac{r\,\frac{\partial}{\partial r} V(t,z,r,\phi)}{2\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial r)^2} V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial}{\partial r} V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)}$$

$$\Lambda = \frac{\frac{\partial}{\partial t}F(t)^2}{12\,F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{6\,F(t)} + \frac{\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)}{6\,F(t)\,V(t,z,r,\phi)} + \frac{\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{12\,V(t,z,r,\phi)} + \frac{r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)} + \frac{r\frac{\partial}{\partial r}V(t,z,r,\phi)}{6\,F(t)^2\,V(t,z,r,\phi)}$$
$$+ \frac{1}{12\,F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,rF(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2}V(t,z,r,\phi)}{12\,r^2F(t)^2\,V(t,z,r,\phi)}$$

Weyl Tensor Components

$$\Psi_0 = -\frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{2\,rF(t)^2\,V(t,z,r,\phi)} + \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{2\,r^2F(t)^2\,V(t,z,r,\phi)}$$

$$\Psi_1 = \frac{\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} + \frac{i\,\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)} - \frac{i\,\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)}$$

$$\Psi_2 = 0$$

$$\Psi_3 = \frac{\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{i\,\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{4\,rF(t)\,V(t,z,r,\phi)} + \frac{i\,\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)}$$

$$\Psi_4 = \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{2\,rF(t)^2\,V(t,z,r,\phi)} - \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{2\,r^2F(t)^2\,V(t,z,r,\phi)}$$

Newman-Penrose Equations

$$NP1 = \frac{\frac{\partial}{\partial t}F(t)^2}{2\,F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{2\,F(t)} + \frac{\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)}{2\,F(t)\,V(t,z,r,\phi)} + \frac{r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{2\,F(t)^2\,V(t,z,r,\phi)} + \frac{3\,r\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{2\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,rF(t)^2\,V(t,z,r,\phi)} - \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)}$$

$$NP2 = \frac{\frac{\partial}{\partial t}F(t)^2}{2\,F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{2\,F(t)} + \frac{\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)}{2\,F(t)\,V(t,z,r,\phi)} + \frac{r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} + \frac{r\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} + \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)}$$

$$NP4 = -\frac{\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{2\,F(t)^2\,V(t,z,r,\phi)}$$

$$NP5 = -\frac{\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{2\,F(t)^2\,V(t,z,r,\phi)}$$

$$\begin{aligned}NP6 = {} & \frac{\frac{\partial}{\partial t}F(t)^2}{24\,F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{6\,F(t)} - \frac{\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)}{6\,F(t)\,V(t,z,r,\phi)} + \frac{7\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{24\,V(t,z,r,\phi)} - \frac{5\,r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)} - \frac{r\frac{\partial}{\partial r}V(t,z,r,\phi)}{6\,F(t)^2\,V(t,z,r,\phi)} \\ & + \frac{1}{24\,F(t)^2} + \frac{5\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)}\end{aligned}$$

$$\begin{aligned}NP7 = {} & -\frac{3\frac{\partial}{\partial t}F(t)^2}{4\,F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{2\,F(t)} + \frac{\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)}{2\,F(t)\,V(t,z,r,\phi)} + \frac{\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{4\,V(t,z,r,\phi)} + \frac{r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} \\ & + \frac{r\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{1}{4\,F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)}\end{aligned}$$

$$\begin{aligned}NP8 = {} & -\frac{\frac{\partial}{\partial t}F(t)^2}{3\,F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{3\,F(t)} + \frac{\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)}{3\,F(t)\,V(t,z,r,\phi)} + \frac{\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{6\,V(t,z,r,\phi)} + \frac{5\,r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)} + \frac{7\,r\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)} \\ & + \frac{1}{6\,F(t)^2} - \frac{5\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)} - \frac{\frac{\partial}{\partial r}V(t,z,r,\phi)}{6\,rF(t)^2\,V(t,z,r,\phi)} + \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)} + \frac{\frac{\partial^2}{(\partial \phi)^2}V(t,z,r,\phi)}{12\,r^2F(t)^2\,V(t,z,r,\phi)}\end{aligned}$$

$$\begin{aligned}NP10 = {} & -\frac{\frac{\partial}{\partial t}F(t)^2}{2\,F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{2\,F(t)} - \frac{\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)}{2\,F(t)\,V(t,z,r,\phi)} - \frac{r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} - \frac{r\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} \\ & + \frac{\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} + \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)} + \frac{\frac{\partial^2}{(\partial \phi)^2}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)}\end{aligned}$$

$$\begin{aligned}NP12 = {} & \frac{17\frac{\partial}{\partial t}F(t)^2}{24\,F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{6\,F(t)} + \frac{\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)}{6\,F(t)\,V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{24\,V(t,z,r,\phi)} - \frac{r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)} \\ & + \frac{r\frac{\partial}{\partial r}V(t,z,r,\phi)}{6\,F(t)^2\,V(t,z,r,\phi)} + \frac{5}{24\,F(t)^2} + \frac{\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)} - \frac{5\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)} - \frac{5\frac{\partial^2}{(\partial \phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)}\end{aligned}$$

$$NP14 = \frac{\frac{\partial}{\partial t}F(t)^2}{2F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{2F(t)} + \frac{\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)}{2F(t)V(t,z,r,\phi)} + \frac{r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{2F(t)^2V(t,z,r,\phi)} + \frac{3r\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)}$$
$$- \frac{\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{2F(t)^2V(t,z,r,\phi)} - \frac{\frac{\partial}{\partial r}V(t,z,r,\phi)}{4rF(t)^2V(t,z,r,\phi)} + \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4r^2F(t)^2V(t,z,r,\phi)}$$

$$NP15 = -\frac{\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{2F(t)^2V(t,z,r,\phi)}$$

$$NP16 = -\frac{3\frac{\partial}{\partial t}F(t)^2}{4F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{2F(t)} + \frac{\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)}{2F(t)V(t,z,r,\phi)} + \frac{\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{4V(t,z,r,\phi)} + \frac{r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} + \frac{r\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)}$$
$$- \frac{1}{4F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} + \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4r^2F(t)^2V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2}V(t,z,r,\phi)}{4r^2F(t)^2V(t,z,r,\phi)}$$

$$NP17 = \frac{\frac{\partial}{\partial t}F(t)^2}{3F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2}F(t)}{3F(t)} - \frac{\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)}{3F(t)V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{6V(t,z,r,\phi)} - \frac{5r^2\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{12F(t)^2V(t,z,r,\phi)} - \frac{7r\frac{\partial}{\partial r}V(t,z,r,\phi)}{12F(t)^2V(t,z,r,\phi)}$$
$$- \frac{1}{6F(t)^2} + \frac{5\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{12F(t)^2V(t,z,r,\phi)} + \frac{\frac{\partial}{\partial r}V(t,z,r,\phi)}{6rF(t)^2V(t,z,r,\phi)} + \frac{i\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial \phi}V(t,z,r,\phi)}{4r^2F(t)^2V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2}V(t,z,r,\phi)}{12r^2F(t)^2V(t,z,r,\phi)}$$

$$NP18 = \frac{\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{2F(t)^2V(t,z,r,\phi)}$$

Equations are vanished.

Bianchi Identities

$$
\begin{aligned}
BI1 = {} & \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial t)^2}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)^2} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r^2\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r\frac{\partial}{\partial r}V(t,z,r,\phi)^2}{8F(t)^3V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{2F(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{2rF(t)^2V(t,z,r,\phi)} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)} + \frac{3i\sqrt{2}r\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)} - \frac{3i\sqrt{2}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)^3V(t,z,r,\phi)} \\
& - \frac{i\sqrt{2}\frac{\partial^2}{(\partial t)^2}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)^2V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)^2V(t,z,r,\phi)^2} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)^2}{8rF(t)^3V(t,z,r,\phi)^2} \\
& - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{8rF(t)^3V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)} - \frac{3i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{8rF(t)^3V(t,z,r,\phi)} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{\partial r(\partial\phi)^2}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)^2} + \frac{3i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)^2} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{2r^3F(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{8r^3F(t)^3V(t,z,r,\phi)^2} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial\phi)^3}V(t,z,r,\phi)}{8r^3F(t)^3V(t,z,r,\phi)}
\end{aligned}
$$

$$
\begin{aligned}
BI2 = {} & \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^3}{8F(t)^3} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial t}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{8F(t)V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{8V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{8V(t,z,r,\phi)} \\
& + \frac{\sqrt{2}r^2\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial z}V(t,z,r,\phi)}{8V(t,z,r,\phi)^2} + \frac{\sqrt{2}r\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\frac{\partial}{\partial z}V(t,z,r,\phi)}{8V(t,z,r,\phi)^3} \\
& - \frac{\sqrt{2}r\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{2F(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{8F(t)^3} - \frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{2rF(t)^3V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{4r^2F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{4rF(t)^3V(t,z,r,\phi)} - \frac{3i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{4r^2F(t)^3V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}V(t,z,r,\phi)}{4r^2F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{2r^2F(t)^3V(t,z,r,\phi)}
\end{aligned}
$$

$$
\begin{aligned}
BI3 = & -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^3}{8F(t)^3} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial t}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{8F(t)V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{8V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{8V(t,z,r,\phi)} \\
& - \frac{\sqrt{2}r^2\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial z}V(t,z,r,\phi)}{8V(t,z,r,\phi)^2} - \frac{\sqrt{2}r\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\frac{\partial}{\partial z}V(t,z,r,\phi)}{8V(t,z,r,\phi)^3} \\
& + \frac{\sqrt{2}r\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{2F(t)^3V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{8F(t)^3} + \frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{2rF(t)^3V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{4r^2F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{4rF(t)^3V(t,z,r,\phi)} - \frac{3i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{4r^2F(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}V(t,z,r,\phi)}{4r^2F(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{2r^2F(t)^3V(t,z,r,\phi)}
\end{aligned}
$$

$$
\begin{aligned}
BI4 = & -\frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial t)^2}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{8F(t)^2V(t,z,r,\phi)^2} \\
& - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r^2\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r\frac{\partial}{\partial r}V(t,z,r,\phi)^2}{8F(t)^3V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{2F(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{2rF(t)^2V(t,z,r,\phi)} \\
& - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)} + \frac{3i\sqrt{2}r\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)} - \frac{3i\sqrt{2}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)^3V(t,z,r,\phi)} \\
& - \frac{i\sqrt{2}\frac{\partial^2}{(\partial t)^2}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)^2V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)^2V(t,z,r,\phi)^2} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)^2}{8rF(t)^3V(t,z,r,\phi)^2} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{8rF(t)^3V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)} - \frac{3i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{8rF(t)^3V(t,z,r,\phi)} \\
& - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{\partial r(\partial\phi)^2}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8rF(t)^3V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)^2} + \frac{3i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)^2} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{2r^3F(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{8r^3F(t)^3V(t,z,r,\phi)^2} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial\phi)^3}V(t,z,r,\phi)}{8r^3F(t)^3V(t,z,r,\phi)}
\end{aligned}
$$

$$
\begin{aligned}
BI5 = & -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^3}{24\,F(t)^3} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial t)^2}F(t)}{4\,F(t)^2} - \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}F(t)}{12\,F(t)} - \frac{5\sqrt{2}\frac{\partial}{\partial t}F(t)^2\,\frac{\partial}{\partial t}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)}{12\,F(t)\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)^2}{12\,F(t)\,V(t,z,r,\phi)^2} \\
& + \frac{7\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{24\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{12\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{12\,V(t,z,r,\phi)} + \frac{11\sqrt{2}r^2\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)} - \frac{2\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{3\,F(t)^3\,V(t,z,r,\phi)} \\
& + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial z}V(t,z,r,\phi)}{12\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{\partial t\partial z}V(t,z,r,\phi)}{12\,F(t)\,V(t,z,r,\phi)^2} + \frac{2\sqrt{2}r\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{3\,F(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial z}V(t,z,r,\phi)}{12\,F(t)\,V(t,z,r,\phi)^3} \\
& - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial z}V(t,z,r,\phi)}{12\,V(t,z,r,\phi)^3} - \frac{\sqrt{2}r^2\frac{\partial^3}{\partial z(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} - \frac{13\sqrt{2}r\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,F(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}r\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)^2} \\
& + \frac{\sqrt{2}r^2\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^3} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{24\,F(t)^3} - \frac{11\sqrt{2}\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)} - \frac{\sqrt{2}r\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}r\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)^3} \\
& + \frac{2\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{3\,F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} + \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{2\,rF(t)^3\,V(t,z,r,\phi)} - \frac{5\sqrt{2}\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)} \\
& - \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial^3}{\partial z(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} + \frac{5\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,rF(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^3} \\
& - \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{4\,r^2F(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)^3} \\
& + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{6\,r^2F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial^3}{\partial z(\partial\phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)^3}
\end{aligned}
$$

$$
\begin{aligned}
BI6 = & -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^3}{24\,F(t)^3} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial t)^2}F(t)}{4\,F(t)^2} - \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}F(t)}{12\,F(t)} - \frac{5\sqrt{2}\frac{\partial}{\partial t}F(t)^2\,\frac{\partial}{\partial t}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)}{12\,F(t)\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)^2}{12\,F(t)\,V(t,z,r,\phi)^2} \\
& + \frac{7\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{24\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{12\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{12\,V(t,z,r,\phi)} + \frac{11\sqrt{2}r^2\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)} - \frac{2\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{3\,F(t)^3\,V(t,z,r,\phi)} \\
& + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial z}V(t,z,r,\phi)}{12\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{\partial t\partial z}V(t,z,r,\phi)}{12\,F(t)\,V(t,z,r,\phi)^2} + \frac{2\sqrt{2}r\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{3\,F(t)^2\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial z}V(t,z,r,\phi)}{12\,F(t)\,V(t,z,r,\phi)^3} \\
& + \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial z}V(t,z,r,\phi)}{12\,V(t,z,r,\phi)^3} + \frac{\sqrt{2}r^2\frac{\partial^3}{\partial z(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} - \frac{13\sqrt{2}r\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,F(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}r\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)^2} \\
& - \frac{\sqrt{2}r^2\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^3} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{24\,F(t)^3} - \frac{11\sqrt{2}\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}r\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}r\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)^3} \\
& + \frac{2\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{3\,F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} - \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{2\,rF(t)^3\,V(t,z,r,\phi)} - \frac{5\sqrt{2}\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{4\,r^2F(t)^2\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial^3}{\partial z(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} + \frac{5\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,rF(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^3} \\
& + \frac{i\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{4\,r^2F(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)^3} \\
& + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{6\,r^2F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial^3}{\partial z(\partial\phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{24\,r^2F(t)^2\,V(t,z,r,\phi)^3}
\end{aligned}
$$

$$
\begin{aligned}
BI7 = & -\frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{(\partial t)^2\partial r}V(t,z,r,\phi)}{24\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)^2\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial t)^2}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{8\,F(t)^2\,V(t,z,r,\phi)} \\
& - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,F(t)\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r^2\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^3\,V(t,z,r,\phi)^2} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r^2\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{12\,F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r\frac{\partial}{\partial r}V(t,z,r,\phi)^2}{24\,F(t)^3\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{3\,F(t)^3\,V(t,z,r,\phi)} - \frac{5i\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial\phi}V(t,z,r,\phi)}{24\,rF(t)\,V(t,z,r,\phi)} \\
& - \frac{5i\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{12\,rF(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{6\,F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^3\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{12\,F(t)^3\,V(t,z,r,\phi)} \\
& - \frac{i\sqrt{2}r\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{3\,F(t)^3\,V(t,z,r,\phi)} + \frac{3i\sqrt{2}\frac{\partial}{\partial t}F(t)^2\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8\,rF(t)^3\,V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial t)^2}F(t)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8\,rF(t)^2\,V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)^2} \\
& + \frac{i\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{12\,rF(t)\,V(t,z,r,\phi)^2} - \frac{i\sqrt{2}r\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{24\,F(t)^3\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)^2}{12\,rF(t)^3\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{12\,rF(t)^3\,V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{12\,F(t)^3\,V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{24\,F(t)^3\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,r^2F(t)^3\,V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{3\,rF(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{\partial r(\partial\phi)^2}V(t,z,r,\phi)}{6\,r^2F(t)^3\,V(t,z,r,\phi)} \\
& - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8\,rF(t)^3\,V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{24\,rF(t)^3\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{12\,r^2F(t)^3\,V(t,z,r,\phi)^2} - \frac{5i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{12\,r^2F(t)^3\,V(t,z,r,\phi)} \\
& - \frac{i\sqrt{2}\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{12\,r^2F(t)^3\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{12\,r^3F(t)^3\,V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{2\,r^3F(t)^3\,V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{12\,r^3F(t)^3\,V(t,z,r,\phi)^2} + \frac{i\sqrt{2}\frac{\partial^3}{(\partial\phi)^3}V(t,z,r,\phi)}{12\,r^3F(t)^3\,V(t,z,r,\phi)}
\end{aligned}
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\begin{aligned}
BI8 = & -\frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{(\partial t)^2\partial r}V(t,z,r,\phi)}{24\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{12\,F(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)^2\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial t)^2}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{8\,F(t)^2\,V(t,z,r,\phi)} \\
& - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,F(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{24\,F(t)\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r^2\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^3\,V(t,z,r,\phi)^2} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r^2\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{12\,F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r\frac{\partial}{\partial r}V(t,z,r,\phi)^2}{24\,F(t)^3\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{3\,F(t)^3\,V(t,z,r,\phi)} + \frac{5i\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial\phi}V(t,z,r,\phi)}{24\,rF(t)\,V(t,z,r,\phi)} \\
& + \frac{5i\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{12\,rF(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{6\,F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{24\,F(t)^3\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{12\,F(t)^3\,V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}r\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{3\,F(t)^3\,V(t,z,r,\phi)} - \frac{3i\sqrt{2}\frac{\partial}{\partial t}F(t)^2\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8\,rF(t)^3\,V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial^2}{(\partial t)^2}F(t)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8\,rF(t)^2\,V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{24\,rF(t)^2\,V(t,z,r,\phi)^2} \\
& - \frac{i\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{12\,rF(t)\,V(t,z,r,\phi)^2} + \frac{i\sqrt{2}r\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{24\,F(t)^3\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)^2}{12\,rF(t)^3\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{12\,rF(t)^3\,V(t,z,r,\phi)} \\
& - \frac{i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{12\,F(t)^3\,V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{24\,F(t)^3\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{12\,r^2F(t)^3\,V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{3\,rF(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{\partial r(\partial\phi)^2}V(t,z,r,\phi)}{6\,r^2F(t)^3\,V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{8\,rF(t)^3\,V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{24\,rF(t)^3\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{12\,r^2F(t)^3\,V(t,z,r,\phi)^2} + \frac{5i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{12\,r^2F(t)^3\,V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\frac{\partial}{\partial r}V(t,z,r,\phi)\,\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{12\,r^2F(t)^3\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{12\,r^3F(t)^3\,V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{2\,r^3F(t)^3\,V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{12\,r^3F(t)^3\,V(t,z,r,\phi)^2} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial\phi)^3}V(t,z,r,\phi)}{12\,r^3F(t)^3\,V(t,z,r,\phi)}
\end{aligned}
$$

$$
\begin{aligned}
BI9 = & -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^3}{8\,F(t)^3} + \frac{3\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial t)^2}F(t)}{8\,F(t)^2} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^2\,\frac{\partial}{\partial t}V(t,z,r,\phi)}{2\,F(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{4\,F(t)\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{16\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{16\,V(t,z,r,\phi)} \\
& + \frac{7\sqrt{2}r^2\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{16\,F(t)^2\,V(t,z,r,\phi)} - \frac{5\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{8\,F(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16\,F(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial z}V(t,z,r,\phi)}{16\,V(t,z,r,\phi)^2} + \frac{3\sqrt{2}r\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{4\,F(t)^2\,V(t,z,r,\phi)} \\
& - \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\,\frac{\partial}{\partial z}V(t,z,r,\phi)}{16\,V(t,z,r,\phi)^3} - \frac{\sqrt{2}r^2\frac{\partial^3}{\partial z(\partial r)^2}V(t,z,r,\phi)}{16\,F(t)^2\,V(t,z,r,\phi)^2} - \frac{5\sqrt{2}r\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{4\,F(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}r^2\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16\,F(t)^2\,V(t,z,r,\phi)^3} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{8\,F(t)^3} \\
& - \frac{7\sqrt{2}\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{16\,F(t)^2\,V(t,z,r,\phi)} + \frac{5\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{8\,F(t)^3\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16\,F(t)^2\,V(t,z,r,\phi)^2} - \frac{5\sqrt{2}\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{16\,rF(t)^2\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial^3}{\partial z(\partial r)^2}V(t,z,r,\phi)}{16\,F(t)^2\,V(t,z,r,\phi)^2} \\
& + \frac{5\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{8\,rF(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{16\,rF(t)^2\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16\,F(t)^2\,V(t,z,r,\phi)^3} - \frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}V(t,z,r,\phi)}{16\,r^2F(t)^2\,V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{16\,rF(t)^2\,V(t,z,r,\phi)^2} \\
& + \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial}{\partial r}V(t,z,r,\phi)}{16\,rF(t)^2\,V(t,z,r,\phi)^3} + \frac{3\sqrt{2}\frac{\partial}{\partial t}F(t)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{8\,r^2F(t)^3\,V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{16\,r^2F(t)^2\,V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial^3}{\partial z(\partial\phi)^2}V(t,z,r,\phi)}{16\,r^2F(t)^2\,V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\,\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{16\,r^2F(t)^2\,V(t,z,r,\phi)^3}
\end{aligned}
$$

$$
\begin{aligned}
BI10 = {} & \frac{3\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{(\partial t)^2\partial r}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{F(t)^2V(t,z,r,\phi)} + \frac{3\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16F(t)V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}r^2\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)^2} - \frac{3\sqrt{2}\sqrt{r+1}\sqrt{r-1}r^2\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)} \\
& - \frac{5\sqrt{2}\sqrt{r+1}\sqrt{r-1}r\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial\phi}V(t,z,r,\phi)}{16rF(t)V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi)}{2rF(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)^2} + \frac{3\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{(\partial r)^3}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)} - \frac{7i\sqrt{2}r\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{2rF(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{16rF(t)V(t,z,r,\phi)^2} - \frac{i\sqrt{2}r\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{16F(t)^3V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)^2}{16rF(t)^3V(t,z,r,\phi)^2} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16rF(t)^3V(t,z,r,\phi)} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)}{16r^2F(t)^3V(t,z,r,\phi)} + \frac{7i\sqrt{2}\frac{\partial^3}{(\partial r)^2\partial\phi}V(t,z,r,\phi)}{16rF(t)^3V(t,z,r,\phi)} - \frac{3\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^3}{\partial r(\partial\phi)^2}V(t,z,r,\phi)}{16r^2F(t)^3V(t,z,r,\phi)} \\
& + \frac{i\sqrt{2}\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{16rF(t)^3V(t,z,r,\phi)^2} - \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{16r^2F(t)^3V(t,z,r,\phi)^2} - \frac{7i\sqrt{2}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi)}{16r^2F(t)^3V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial r}V(t,z,r,\phi)\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{16r^2F(t)^3V(t,z,r,\phi)^2} \\
& + \frac{\sqrt{2}\sqrt{r+1}\sqrt{r-1}\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{8r^3F(t)^3V(t,z,r,\phi)} + \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)}{2r^3F(t)^3V(t,z,r,\phi)} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{16r^3F(t)^3V(t,z,r,\phi)^2} + \frac{i\sqrt{2}\frac{\partial^3}{(\partial\phi)^3}V(t,z,r,\phi)}{16r^3F(t)^3V(t,z,r,\phi)}
\end{aligned}
$$

$$
\begin{aligned}
BI11 = {} & -\frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^3}{8F(t)^3} + \frac{3\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}F(t)}{8F(t)^2} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)^2\frac{\partial}{\partial t}V(t,z,r,\phi)}{2F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{4F(t)V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)}{16V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^3}V(t,z,r,\phi)}{16V(t,z,r,\phi)} \\
& + \frac{7\sqrt{2}r^2\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)} - \frac{5\sqrt{2}r^2\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}r^2\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial^3}{(\partial t)^2\partial z}V(t,z,r,\phi)}{16V(t,z,r,\phi)^2} + \frac{3\sqrt{2}r\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} \\
& + \frac{\sqrt{2}\frac{\partial^2}{(\partial t)^2}V(t,z,r,\phi)\frac{\partial}{\partial z}V(t,z,r,\phi)}{16V(t,z,r,\phi)^3} + \frac{\sqrt{2}r^2\frac{\partial^3}{\partial z(\partial r)^2}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2} - \frac{5\sqrt{2}r\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^3V(t,z,r,\phi)} - \frac{\sqrt{2}r^2\frac{\partial}{\partial z}V(t,z,r,\phi)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^3} - \frac{\sqrt{2}\frac{\partial}{\partial t}F(t)}{8F(t)^3} \\
& - \frac{7\sqrt{2}\frac{\partial^3}{\partial t(\partial r)^2}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)} + \frac{5\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{8F(t)^3V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2} - \frac{5\sqrt{2}\frac{\partial^2}{\partial t\partial r}V(t,z,r,\phi)}{16rF(t)^2V(t,z,r,\phi)} - \frac{\sqrt{2}\frac{\partial^3}{\partial z(\partial r)^2}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^2} \\
& + \frac{5\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial r}V(t,z,r,\phi)}{8rF(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16rF(t)^2V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\frac{\partial^2}{(\partial r)^2}V(t,z,r,\phi)}{16F(t)^2V(t,z,r,\phi)^3} - \frac{\sqrt{2}\frac{\partial^3}{\partial t(\partial\phi)^2}V(t,z,r,\phi)}{16r^2F(t)^2V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial^2}{\partial z\partial r}V(t,z,r,\phi)}{16rF(t)^2V(t,z,r,\phi)^2} \\
& - \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\frac{\partial}{\partial r}V(t,z,r,\phi)}{16rF(t)^2V(t,z,r,\phi)^3} + \frac{3\sqrt{2}\frac{\partial}{\partial t}F(t)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{8r^2F(t)^3V(t,z,r,\phi)} + \frac{\sqrt{2}\frac{\partial}{\partial t}V(t,z,r,\phi)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{16r^2F(t)^2V(t,z,r,\phi)^2} + \frac{\sqrt{2}\frac{\partial^3}{\partial z(\partial\phi)^2}V(t,z,r,\phi)}{16r^2F(t)^2V(t,z,r,\phi)^2} - \frac{\sqrt{2}\frac{\partial}{\partial z}V(t,z,r,\phi)\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{16r^2F(t)^2V(t,z,r,\phi)^3}
\end{aligned}
$$

Equations are not vanished.

Petrov Type

Type "I"